Politely — Rewriting Impolite Sentences Politely with an Encoder–Decoder Built from Scratch

NLP course project — Text generation with encoder–decoder architectures

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Every number in this report was produced by the code in this repository and can be found in `reports/results.json`, `reports/dataset_stats.json`, `reports/vocab_stats.json` and `experiments/<model>/`. Nothing was estimated or copied from elsewhere.

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1. Introduction

1.1 Task

Politeness transfer: given a blunt, impatient or dismissive English sentence, generate a polite sentence that says *the same thing*.

Input (impolite)	Target (polite)
Send me the report now.	Could you please send me the report as soon as possible?
Your poster is full of typos.	Thank you for the poster. I think it has a few typos, though.
Stop guessing, the store closes at 6 pm.	If I remember correctly, the store closes at 6 pm.

1.2 Motivation

Rephrasing a message so that it is received well is something people do every day (emails, customer service, team chats). It is also an ideal problem to study an encoder–decoder: the output depends on the whole input (who, what, when must be kept), but its *form* must change (imperative → question, criticism → softened feedback, refusal → apology). A model that simply copies the input fails, and a model that ignores the input and writes a generic polite sentence fails too.

1.3 Problem formulation

Input: a sequence of tokens $x = (x_1 \dots x_n)$. Output: a sequence $y = (y_1 \dots y_m)$ ending with `<EOS>`. The model learns $P(y \mid x) = \Pi_t P(y_t \mid y_{<t}, x)$ and is trained to minimise the token-level cross-entropy on (impolite, polite) pairs. It is a **sequence-to-sequence paraphrase / style-transfer** problem. A good output must satisfy three conditions at the same time: (1) be polite, (2) keep all the content of the input, (3) be fluent. Section 7 measures each one separately.

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2. Related concepts

Text style transfer changes an attribute of a text (formality, sentiment, politeness) while keeping its content. With parallel data (pairs of sentences), it is treated like translation from "impolite English" to "polite English".

Encoder–decoder (seq2seq). An *encoder* reads the input and turns it into vectors; a *decoder* generates the output one token at a time, conditioned on these vectors and on the tokens it has already produced (Sutskever et al., 2014; Cho et al., 2014).

Recurrent cells. The LSTM (and the lighter GRU) keep a hidden state updated at every token; their gates let information survive across many steps, which plain RNNs cannot do because of vanishing gradients.

Attention (Bahdanau et al., 2015). Without attention, the whole input must fit into one fixed-size vector. Attention lets the decoder look back at *every* encoder state at each step and build a context vector focused on the words that matter for the next output word.

Copy mechanism (pointer-generator, See et al., 2017). At each step the model mixes two distributions: generating a word from the vocabulary, or copying a word of the input (with the attention weights as copy probabilities). It is useful whenever the output repeats input words — exactly the case here, where names, objects, numbers and dates must be carried over.

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3. Dataset

3.1 Source

Public parallel corpora close to this task (detoxification corpora such as ParaDetox) are built from offensive messages and contain insults and swear words, which this project wants to exclude. The dataset was therefore **built for the project**: a reproducible program, `data/generate_dataset.py`, assembles sentence pairs from building blocks written by hand in `data/resources.py`. This gives full control over the content: balanced categories, the same facts on both sides of a pair, no offensive language, and test sets with phrasings never seen in training. The building blocks are:

- **9 categories** of everyday messages, balanced: request, reminder, refusal, criticism, complaint, disagreement, question, stop (asking someone to stop a behaviour), announcement;
- **slot values** (people, documents, objects, rooms, events, times, bills, facilities, services and their problems, facts, questions…) — about 1,800 distinct words;
- per category, **12 impolite phrasings** (templates) and **6 polite phrasings**;
- optional **explanation sentences** ("We are already late.") that must be carried over.

For one input, the generator draws a *content* (e.g. action = *close*, object = *the kitchen window*, time = *now*, reason = *It's raining outside*), realises it with one impolite phrasing, and with three different polite phrasings. Rudeness is carried by **tone only** (orders, impatience, dismissive remarks, harsh judgements of a piece of work); every sentence is checked against a list of banned words (insults, swear words, hateful terms) and none appears.

3.2 Improving the dataset (version 1 → version 2)

The dataset was improved after a first round of experiments. The first version (v1) had two problems:

- **errors and incoherent pairs** — some combinations of building blocks produced sentences that make no sense or are ungrammatical: *“Move your car out of the library”*, *“Give a talk at the football match”*, or *“You were supposed to call Sarah back. **Call it** today.”* (the template “ it” does not work with phrasal verbs or with a person as object);
- **a vocabulary that was too narrow** — office life only, 1,371 words.

The models trained on v1 were very good on v1’s own test set but very bad on sentences written by hand (0 %–1 % success, section 7.4): they recognised known word combinations instead of learning to *keep whatever the input says*.

Version 2 (the final one) keeps the same design, categories and phrasings, and improves the content: incoherent combinations were removed (no cars in libraries; criticism contexts that fit the kind of work — food, art, music, documents), templates such as “ it” are no longer combined with actions where they are ungrammatical, and the vocabulary was widened to everyday life — home, shopping, transport, school, leisure, phrasal verbs (*turn the music down*), explanation sentences — with 14,400 training inputs instead of 10,800. The v1 results are kept in `archive_v1/` and compared in section 7.4.

3.3 Fields and files

`data/politeness_dataset.csv` (and the same rows split into `train.csv`, `valid.csv`, `test.csv`, `test_unseen.csv`):

Column	Meaning
`id`	input id (`req-train-0042`); the references of one input share it
`split`	train / valid / test / test_unseen
`category`	one of the 9 categories
`template_id`	impolite phrasing used (`req_02`); `*_10…12` are held out
`ref_index`	which polite reference (0, 1, 2)
`impolite`	the input
`polite`	a reference output

`data/challenge.csv` contains **72 pairs written by hand** (8 per category), independently of the generator, with other words and structures (“Turn the music down.”, “Where’s the bathroom?”, “Prices go up next month, like it or not.”).

3.4 Statistics (`reports/dataset_stats.json`)

Split	Inputs	Pairs	Refs / input	Impolite phrasings	Mean length in / out (tokens)
train	14,400	28,800	2	81 (seen)	13.1 / 17.8
valid	1,350	4,050	3	81 (seen)	13.3 / 18.0
test	1,350	4,050	3	81 (seen)	13.2 / 18.0
test_unseen	900	2,700	3	27 (**never seen**)	14.1 / 17.9
challenge	72	72	1	hand-written	8.0 / 11.9

- Every category has exactly 1,600 / 150 / 150 / 100 inputs in train / valid / test / test_unseen (figure `dataset_categories.png`).
- **0** input shared between two splits, **0** duplicate pair, **0** pair where output = input.
- 52.8 % of the target tokens already appear in the input (figure `dataset_lengths_overlap.png`): half of the job is copying, half is rewriting.
- Limitation: distinct-3 (share of distinct word trigrams) is 0.18 in train — the phrasings come from a finite set of templates.

3.5 Split

The split is made **by content**, not by pair: a content (action + object + time + reason…) used in one split is never used in another, and all references of one input stay in the same split. A test sentence is therefore never a training sentence with a different wording. Three test sets measure three levels of generalisation:

1. **test** — new content, phrasings seen in training;
2. **test_unseen** — new content *and* impolite phrasings never seen in training (3 of the 12 phrasings of each category are reserved for it);
3. **challenge** — hand-written sentences, outside the generator.

Validation is used for every choice (early stopping, model, decoding); the test sets are measured once, at the end.

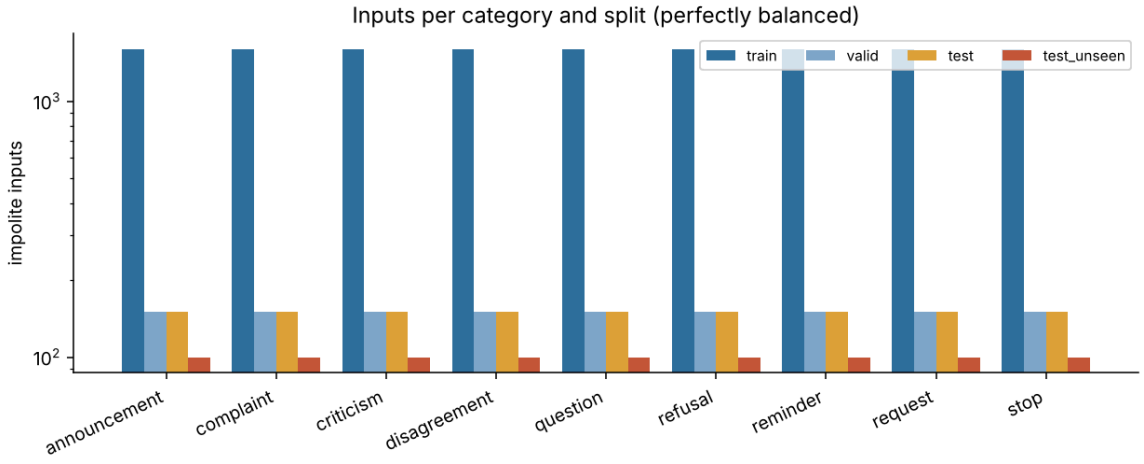


Figure 1 — Inputs per category and split.

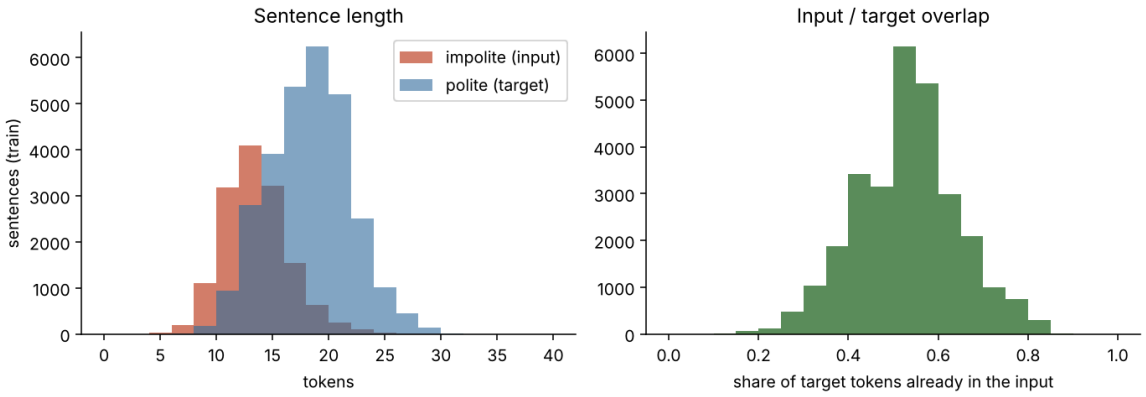


Figure 2 — Sentence lengths and input/target overlap (train).

4. Preprocessing (`preprocessing.py`)

Step	What	Why, for this task
Normalisation	Unicode NFKC, curly quotes → straight quotes, spaces collapsed	the same word must always give the same token (`don’t` = `don't`)
Lowercasing	yes	politeness does not depend on case; halves the number of word forms. The interface restores the user’s capitals for display (`TV`, `Sarah`).
Punctuation	each mark is a separate token	`?` versus `.` carries meaning here: a polite request is usually a question
Tokenisation	regex `\w+(?:'\w+)?` or one punctuation mark; contractions stay one token (`don't`, `i'm`)	word level is enough for short sentences and keeps sequences short (13 tokens on average)
Vocabulary	built on the **training split only**; words seen at least twice	rare words would get untrained embeddings; building on train only avoids leakage
Special tokens	`<PAD>`=0, `<SOS>`=1, `<EOS>`=2, `<UNK>`=3	padding, start and end of generation, unknown words
Padding	batches are padded to the longest sentence with `<PAD>`; the encoder uses packed sequences and attention uses a mask, so padding is never read	variable-length sentences in one tensor
Truncation	inputs and targets cut at 40 tokens (the longest sentence has 31)	bounds memory and decoding time; also the limit enforced by the interface
Unknown words	mapped to `<UNK>`; the copy model gives each unknown input word a temporary id so it can still be copied	names, numbers and rare objects are frequent here

Vocabulary size: 1,805 (1,801 words + 4 special tokens), shared by the encoder and the decoder because input and output are the same language. Proportion of unknown tokens (`reports/vocab_stats.json`): 0 % in train, 0.5 % in valid/test, 4.7 % in test_unseen, 11.5 % in challenge (19.3 % with the v1 vocabulary).

Numerical representation. `"Close the window now."` → `[close, the, window, now, .]` → `[ids…, EOS]` → each id is turned into a **128-dimensional embedding** by `nn.Embedding` (trained from scratch, `padding_idx=0`). The decoder input is `<SOS>` + target, the training label is target + `<EOS>`.

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5. Proposed architecture (`model.py`)

5.1 Encoder

`Embedding(1805, 128)` → dropout 0.3 → **LSTM, 1 layer, hidden size 256, unidirectional**. It returns one hidden state per input word (`[B, n, 256]`, used by attention) and the final state `(h_n, c_n)`, which **initialises the decoder**. (A bidirectional option exists in the code, with a linear "bridge" that merges the two directions.)

5.2 Decoder (one step)

At step *t* the decoder receives the **previous token** (the true one during training with teacher forcing, its own prediction at inference) and its previous state:

1. `e = Embedding(y_{t-1})` (same embedding matrix as the encoder);
2. attention: context vector `c_t` (5.3);
3. `s_t = LSTM([e ; c_t], s_{t-1})`;
4. output layer: `[s_t ; c_t ; e]` → linear 128 → dropout → linear 1805 = logits (factorised through 128 dimensions to save parameters);
5. `softmax` → probability of every vocabulary word.

Generation starts with `<SOS>` and **stops when `<EOS>` is produced** or after 40 tokens.

5.3 Attention (Bahdanau, additive)

Why: without attention (model A) the decoder only sees the final encoder state: one vector of 256 numbers must contain every detail of the sentence (the person, the object, the day…). With 13-token inputs full of details, the decoder forgets or mixes them up (section 8).

How: for each encoder state `h_j` and the current decoder state `s`:

`score_j = v^T tanh(W_enc h_j + W_dec s)` (attention dimension 128) → padding positions are masked (score = $-\infty$) → `α = softmax(score)` (weights summing to 1) → **context** `c_t = Σ_j α_j h_j`, concatenated to the decoder input and to the output layer.

The weights `α` are what the interface draws as a heat map.

5.4 Copy mechanism (pointer-generator)

A switch `p_gen = σ(w · [c_t ; s_t ; e] + b)` decides between generating and copying:

```
P(w) = p_gen · P_vocab(w) + (1 − p_gen) · Σ_{j : x_j = w} α_j
```

Unknown input words receive temporary ids beyond the vocabulary, so a word never seen in training (*Grzegorz*, *4721*, *zucchini*) can still be produced. The training targets use the same

extended ids, so the loss teaches the model when to copy.

Word dropout (`unk_dropout = 0.1`, our addition for v2): during training, 10 % of the input words are shown to the encoder as `<UNK>` while their real id is kept for copying. Without it, the training data contain almost no unknown words (0 % of train tokens), so the copy switch never learns what to do with them; with it, the model learns "if I cannot read a word, copy it". Model `C_copy_nodrop` is the ablation.

5.5 Models compared and number of parameters

Model	Change	Parameters
A_lstm	baseline: LSTM encoder–decoder, no attention	1,303,693
B_attention	A + Bahdanau attention	1,664,397
C_copy	B + copy mechanism + word dropout 0.1	1,665,038
C_copy_nodrop	C without word dropout (ablation)	1,665,038
D_gru_attention	B with GRU cells instead of LSTM	1,401,229
E_attention_h128	B with hidden size 128	875,917
E_attention_h512	B with hidden size 512	4,421,005
F_attention_tf0.5	B trained with teacher forcing ratio 0.5	1,664,397

Breakdown for B: shared embedding 231,040 · encoder LSTM 395,264 · attention 65,792 · decoder LSTM 657,408 · output layer 314,893. The copy switch adds 641 parameters.

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6. Training (`train.py`)

Setting	Value	Reason
Loss	token-level cross-entropy (NLL of the log-probabilities), `<PAD>` ignored	standard objective of sequence generation
Optimizer	Adam, learning rate 1e-3	robust default for RNNs
LR schedule	×0.5 when the validation loss does not improve for 1 epoch	finer steps near convergence
Batch size	64 (shuffled each epoch, fixed seed 42)	
Epochs	at most 30, **early stopping** after 3 epochs without validation improvement; the best checkpoint is kept	prevents overfitting, saves time
Embedding / hidden	128 / 256 (E: 128 and 512)	small vocabulary and short sentences
Layers	1	enough for 13-token sentences (see E for capacity)
Dropout	0.3 on embeddings and before the output layer	regularisation
Gradient clipping	global norm 1.0	prevents exploding gradients in RNNs
Teacher forcing	ratio 1.0 (F: 0.5)	see below
Hardware	2-core CPU, no GPU	

Teacher forcing. During training the decoder normally receives the *true* previous word (ratio 1.0): training is fast and stable. With ratio 0.5 (model F), at each step a coin decides whether the decoder receives the true word or its own prediction, which exposes it to its own mistakes as at inference time (“exposure bias”). Validation loss is always computed with full teacher forcing so that all models are compared on the same basis.

Sanity check. Before every full training, each architecture had to memorise 40 pairs (`--overfit 40`, no dropout, 60 epochs). All eight configurations did (logs in `experiments/sanity_*`): the loss of B fell from 7.49 to 0.07, and every model ended between 0.05 and 0.17 with a token accuracy of 0.96–0.97 (it cannot reach 1.0 because each input has two different polite references). This proves that the forward pass, the loss and the masking are correct.

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7. Experimental results

7.1 Metrics and why

Metric	What it measures	Why it is needed here
BLEU (multi-reference, 1–4-grams, written by hand and checked equal to `sacrebleu`)	n-gram overlap with the 3 references	standard for generation; but many correct polite phrasings exist, so BLEU underestimates good outputs
chrF	character n-gram F-score	more tolerant of small variations (*send / sending*)
Exact match	output identical to one reference	strict; low by nature (6 polite phrasings per category)
Content kept	share of the input's content words (present in the input and in every reference, compared after light stemming so that *close/closing* match) found in the output	a polite sentence that drops "Friday" or "Sarah" is wrong
Polite rate	share of outputs that our politeness classifier judges polite (P ≥ 0.5)	the style must actually change
Success	polite **and** all content words kept **and** no repeated word	one number for "usable rewrite" — the main automatic metric
Repetitions	a word immediately repeated ("the the")	typical seq2seq failure
Token accuracy, perplexity	teacher-forced, on the reference pairs	diagnostics of the language model, not of the generated text
Human evaluation	acceptable / partial / failed, judged sentence by sentence	the automatic metrics cannot judge grammar and meaning (section 7.5)

The **politeness classifier** (`politeness.py`) is a logistic regression on word unigrams and bigrams written from scratch in NumPy (12,680 features), trained on the training split only. It recognises 100 % of the valid/test sentences, 98.3 % (balanced) of test_unseen and 88.9 % of the challenge set (it misses 22 % of the hand-written *polite* sentences, which use rarer markers). All neural models reach a polite rate of ≈ 1.0: they always produce a polite *frame*. The differences between models are therefore in **content** and **fluency**.

7.2 Learning curves (figure `learning_curves.png`)

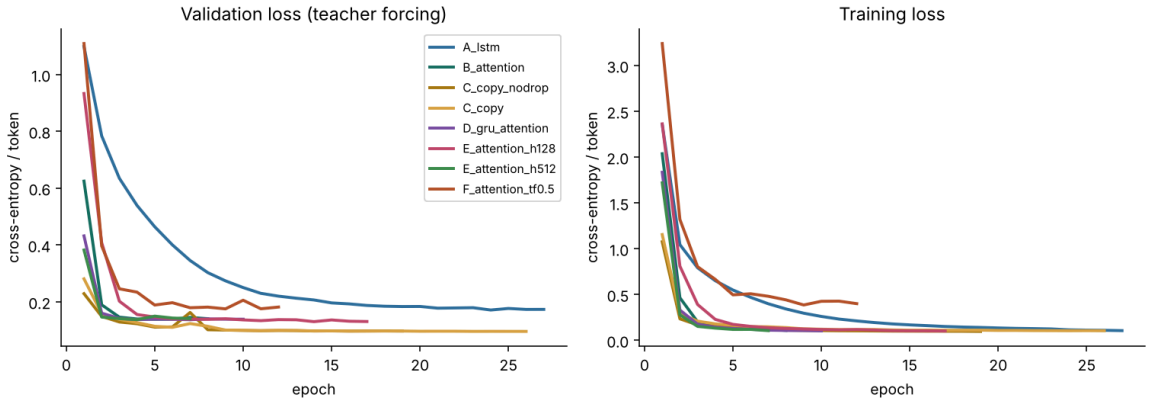


Figure 3 — Validation and training loss of the eight models.

Model	Best epoch / epochs run	Best valid loss	Valid perplexity	Training time
A_lstm	24 / 27	0.171	1.19	27.5 min
B_attention	5 / 8	0.138	1.15	11.9 min
C_copy_nodrop (final)	16 / 19	0.097	1.10	53.7 min*
C_copy (word dropout)	26 / 26	0.096	1.10	39.5 min
D_gru_attention	7 / 10	0.137	1.15	11.9 min
E_attention_h128	14 / 17	0.130	1.14	16.9 min
E_attention_h512	4 / 7	0.137	1.15	21.2 min
F_attention_tf0.5	9 / 12	0.175	1.19	17.9 min

* Its first epoch ran at the same time as other jobs (1,409 s instead of ~90 s); the other 18 epochs took about 28 minutes.

- **Slight overfitting without the copy mechanism, none with it:** for A, the training loss goes below the validation loss from epoch 12 (0.11 vs 0.17 at the best epoch); B, D and E end with a smaller gap (0.01–0.03). The validation loss does not really rise and early stopping keeps the best epoch. With the copy mechanism (C, C'), training and validation losses stay equal (≈ 0.10). Larger models (E_h512) reach their best point earlier and then stall, a first sign that capacity is not the bottleneck.
- **A learns much more slowly** (24 epochs) than the attention models (5–16): squeezing the sentence into one vector is hard.
- The loss cannot reach 0: each input has two different references in training, so the best possible model hesitates between them (the sanity check shows the same ceiling, ≈ 0.97 token accuracy).
- **F (teacher forcing 0.5)** has a higher training loss by construction (it is fed its own mistakes) and a higher validation loss.

7.3 Quantitative evaluation and baseline comparison

Success rate (greedy decoding unless stated; full tables with all metrics in `reports/results.md`):

System	valid	test	test_unseen	challenge	BLEU test	Content kept (test / unseen / challenge)
Copy baseline	0.000	0.000	0.034	0.000	47.5	1.000 / 1.000 / 1.000
Rule-based baseline	0.128	0.125	0.253	0.097	48.2	1.000 / 1.000 / 1.000
A_lstm (no attention)	0.862	0.845	0.350	0.111	74.2	0.961 / 0.584 / 0.343
B_attention	0.921	0.912	0.582	0.222	77.2	0.983 / 0.851 / 0.506
D_gru_attention	0.928	0.928	0.578	0.236	77.3	0.987 / 0.845 / 0.561
E_attention_h128	0.930	0.928	0.630	0.236	77.1	0.987 / 0.875 / 0.516
E_attention_h512	0.923	0.921	0.613	0.236	77.2	0.985 / 0.874 / 0.499
F_attention_tf0.5	0.924	0.921	0.572	0.250	77.2	0.985 / 0.846 / 0.527
C_copy (copy + word dropout)	0.996	0.998	0.682	0.681	78.2	0.999 / 0.868 / 0.889
C_copy_nodrop (copy)	0.998	0.998	0.758	0.764	78.6	0.999 / 0.913 / 0.923
C_copy_nodrop + beam 5 (final)	**1.000**	**0.999**	**0.787**	**0.764**	77.2	1.000 / 0.924 / 0.926

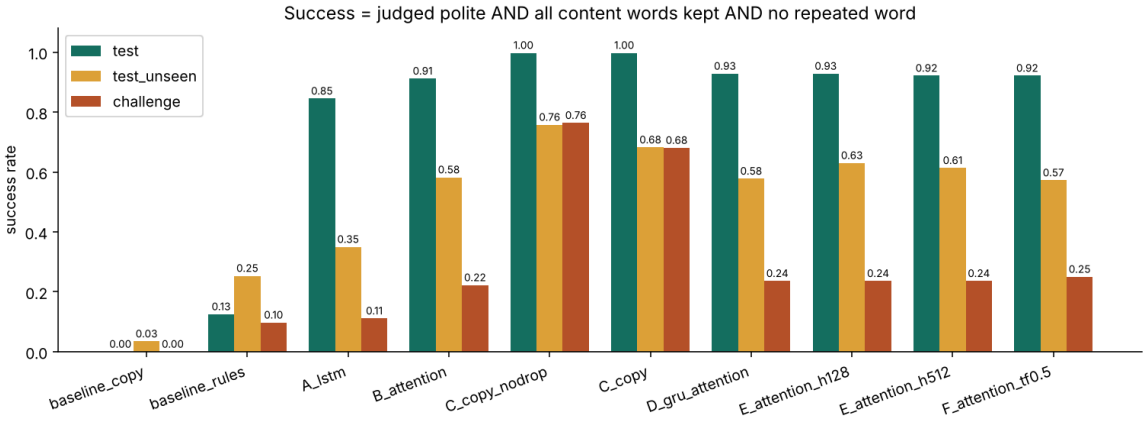


Figure 4 — Success rate of every system on the three test sets.

Choice of the final system. Every choice was made on the validation set. Among the models, C_copy_nodrop has the best validation success (0.998); among its decoding strategies (table below), beam search with 5 beams is the best (1.000). The test sets were then measured once. *Caveat:* the validation set is almost saturated (0.996–1.000 for all copy variants), so it separates these variants only weakly; the generalisation sets tell them apart much more clearly — they happen to agree with the validation choice.

Decoding strategies (C_copy_nodrop, validation, figure `decoding.png`):

Decoding	Success	BLEU	chrF	Exact match	Mean length	Time for 1,350 sentences
greedy	0.998	78.46	83.8	0.512	17.7	1 s (batched)
beam 3	0.999	78.15	83.8	0.518	18.1	27 s
beam 5	**1.000**	78.53	84.2	0.527	18.3	29 s
sampling T = 0.7	0.999	79.34	84.0	0.522	17.9	2.5 s
top-k = 5 (T = 1)	0.998	78.62	83.6	0.511	17.9	2.7 s

Differences are small on this easy set. Beam search never drops content (content kept 1.000) because it compares whole sentences; sampling gives more varied phrasings at a small risk (top-k at T = 1 makes the most mistakes). On test_unseen, beam 5 also beats greedy (0.787 vs 0.758).

7.4 What each change brought (answers to the required questions)

Does the model learn the task? Yes. On new content with known phrasings (test), every neural model is far above both baselines (0.85–1.00 vs 0.00–0.13). The baselines show why the task is not trivial: copying keeps 100 % of the content but is never polite; deleting rude markers and adding “please” is judged polite only 13 % of the time.

Does attention improve the results? Clearly. A → B: success 0.845 → 0.912 on test, 0.350 → 0.582 on test_unseen, 0.111 → 0.222 on challenge; content kept on test_unseen 0.584 → 0.851; validation loss 0.171 → 0.138, reached in 5 epochs instead of 24. Without attention, the model must rebuild every detail from one vector and replaces rare details with frequent ones (section 8).

Does the copy mechanism improve the results? It is the largest single gain: B → C on the hand-written challenge set, success 0.222 → 0.764 and content kept 0.506 → 0.923. Names, numbers and objects can now be copied instead of being generated from the vocabulary. The attention maps show the copy as an almost perfect diagonal on the content part of the sentence (figures `attention_*.png`).

Does increasing model complexity improve performance? No. Hidden size 128 / 256 / 512 (875 k / 1.66 M / 4.42 M parameters) give the same test success (0.928 / 0.912 / 0.921) and similar generalisation (0.630 / 0.582 / 0.613 on test_unseen). The largest model stops improving after 4 epochs. Errors come from *what the model has seen* (data) and *how it can produce words* (copy), not from capacity. The GRU (D, 16 % fewer parameters) is as good as the LSTM.

Teacher forcing. Training on its own predictions half of the time (F) did not help: success 0.921 on test (B: 0.912), 0.572 on test_unseen (B: 0.582), and a higher validation loss. With short sentences and a strong signal, exposure bias is not the main problem here.

Word dropout — a hypothesis that was not confirmed. We expected that hiding 10 % of the input words behind `<UNK>` would teach the copy mechanism to copy unknown words. The opposite happened: C_copy (with word dropout) is worse than C_copy_nodrop on test_unseen (0.682 vs 0.758) and challenge (0.681 vs 0.764) and repeats words more often. A likely

explanation: the pointer-generator already copies unknown words without it, because their extended id makes them copyable, while hiding *known* words removes the information the model needs to decide what kind of sentence it is rewriting. We kept the ablation in the report and chose the model without word dropout.

Effect of the dataset (v1 → v2). The challenge set is the only test set identical in both versions (rescored with the same metric):

	v1 dataset (office only, 1,371 words)	v2 dataset (final, 1,805 words)
A_lstm, challenge success / content kept	0.014 / 0.090	0.111 / 0.343
B_attention, challenge success / content kept	0.000 / 0.191	0.222 / 0.506
Unknown tokens in challenge inputs	19.3 %	11.5 %

Broadening the world of the data mattered as much as the architecture.

7.5 Human evaluation: how far can the automatic metric be trusted?

“Success” checks politeness *markers* and *content words*, not grammar or meaning. To measure the gap, every output of the final system on the challenge set (72), and random samples of 40 outputs on test and on test_unseen, were labelled **acceptable** (polite, same meaning, grammatical), **partial** (meaning recoverable but ungrammatical, a detail lost, or tone not fully softened) or **failed** (meaning changed or garbled). The labels are in `reports/human_eval_*.csv` and the summary in `reports/human_eval_summary.json`. *Note: the labels were produced with the help of an AI assistant; they are the judgement of a single annotator and should be confirmed by more annotators (section 9.3).*

Set (final system)	Automatic success	Human: acceptable	partial	failed
test (40 random)	0.975	**0.975**	0.025	0.000
test_unseen (40 random)	0.775	**0.375**	0.275	0.350
challenge (all 72)	0.764	**0.528**	0.167	0.306
challenge, B_attention (all 72)	0.222	0.167	0.056	0.778

- On sentences like the training data, the automatic metric is reliable (100 % agreement).
- Outside it, the automatic metric is **too optimistic**: it accepts outputs that contain the right words in the wrong grammar (“could you **booked** the restaurant”) or with a reversed meaning (“Would it be possible for you to lend **you** any more money?”).
- The honest headline figure for free-form input is therefore **about half of the rewrites are fully acceptable** (and 70 % are at least understandable) — compared with 17 % for the attention model without copy.

Acceptable rate on the challenge set by category: request 100 %, stop 88 %, disagreement 63 %, complaint 50 %, question 50 %, reminder 38 %, criticism 38 %, announcement 38 %, refusal 13 %.

8. Generated examples

8.1 Successful examples (final system)

Set	Input	Reference (one of 3)	Generated
test	I told you to put the bottles back in the lab. Do it this afternoon. The lab is a mess.	Would it be possible for you to put the bottles back in the lab this afternoon? The lab is a mess.	Could I ask you to put the bottles back in the lab this afternoon? Thank you! The lab is a mess.
test	Again! The package number 7690 was sent to the wrong address last week.	I hope you can help: my package number 7690 was sent to the wrong address last week. Thank you in advance.	Hi, I'm writing because my package number 7690 was sent to the wrong address last week. I would appreciate your help.
test_unseen	How many times? Stop borrowing my bags without asking. I have mentioned it twice already.	—	Could you please avoid borrowing my bags without asking? I have mentioned it twice already.
challenge	Give me the keys to the storage room.	Could you please give me the keys to the storage room?	Could I ask you to give me the keys to the storage room? Thank you!
challenge	Turn the music down.	Could you please turn the music down?	Could I ask you to turn the music down? Thank you!
challenge	My room has no hot water.	Excuse me, there is no hot water in my room. Could you please look into it?	I'm sorry to bother you, but my room has no hot water. Could you help me?
challenge	You're wrong. Water boils at 100 degrees.	Actually, I think water boils at 100 degrees.	Actually, I think water boils at 100 degrees. Could you double-check?
challenge	Which bus goes to the airport?	Excuse me, do you know which bus goes to the airport?	Could you tell me which bus goes to the airport, please?

The model rewrites the *frame* (order → question, criticism → "I appreciate the effort on…, but…", complaint → "I'm sorry to bother you, but…"), removes the rude markers ("Do it", "Again!", "How many times?") and copies the content word for word, including the number 7690 and the explanation sentence.

8.2 Failure cases

Input	Generated	What went wrong
I won’t lend you any more money. *(challenge)*	Would it be possible for you to lend you any more money?	refusal misread as a request: **meaning reversed**
I waited forty minutes for my food. *(challenge)*	I appreciate the effort on the food, but it is a little overcooked.	no known frame fits, the model **invents** a criticism
You were supposed to pay the rent yesterday. *(challenge)*	Whenever you get a chance, could you pay the rent yesterday?	content copied, but the **tense** is not adapted
Where’s the bathroom? *(challenge)*	I would really appreciate it if you could where’s the bathroom.	contraction *where’s* never seen in a question: wrong frame, ungrammatical
How is this acceptable? The fridge has not been working since Thursday. *(test_unseen)*	Could we take another look at the acceptable? It is a little has.	a **rude opener never seen in training** is taken for content
Do I have to remind you every day to lock the garden when you leave? *(test_unseen)*	Could you please remind you every day to lock the garden when you?	the rude part is copied as if it were the request
You still haven’t lent me your remote control … I need it in ten minutes. *(test)*	… before monday? I need it in monday minutes.	a copy pointed at the wrong word (*Monday* instead of *ten*)

The model without attention (A) fails differently: it produces fluent, polite sentences about the **wrong content** — “Give me the keys to the storage room.” → “I would really appreciate it if you could send the newsletter to the shared folder.” (v1) — because the summary vector lost the details.

8.3 Qualitative analysis: why the model fails

1. **Frame selection is learned from surface cues.** The decoder chooses one of the polite frames it knows from a few cue words. When the cues are new (“Not interested in…”, “Prices go up…”, “How is this acceptable?”), it picks a wrong frame and then forces the content into it.
2. **Rude openers it has never seen are treated as content** and copied (“remind you every day”). In training, every word that is not a known rude marker is content to keep.
3. **Grammar is not re-computed.** The copy mechanism copies words as they are; the model rarely learned to change *booked* → *book*, *yesterday* → *soon*, or *where’s* → *where … is*, because the generator’s polite phrasings almost never require it.
4. **Unknown and rare words** are copied correctly most of the time (*Canberra*, *2019*, *voice messages*), which is precisely what the copy mechanism was added for.

Difficult inputs are therefore: refusals and announcements phrased freely (13 % and 38 % acceptable), sentences that need a grammatical change, and rude formulas that do not appear in the training data.

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9. Discussion

9.1 Interpretation

- The task is learned, and the improvements come in a clear order of importance: **copy mechanism > attention > data diversity >> size, cell type, teacher forcing ratio**.
- The three test sets reveal three levels of generalisation: near-perfect on new content (0.999), good on new phrasings (0.787 automatic, 38 % fully acceptable by hand), and partial on free-form sentences (0.764 automatic, 53 % acceptable by hand).
- Automatic metrics agree with humans on familiar inputs and overestimate quality on unfamiliar ones. Reporting only the test-set score (0.999) would have been misleading.

9.2 Strengths

- Complete pipeline written from scratch: tokenizer, vocabulary, encoder, Bahdanau attention, pointer-generator copy, three decoding strategies, all metrics (BLEU checked against sacrebleu), politeness classifier.
- Clean experimental protocol: split by content, held-out phrasings, independent hand-written set, choices on validation only, baselines in every table, sanity check, human evaluation.
- Small and fast: 1.67 M parameters, under 30 minutes of CPU training (19 epochs of ~90 s), about 20 ms per sentence with beam search (29 s for 1,350 sentences).
- Dataset free of insults, balanced, reproducible (deterministic generator).

9.3 Limitations

- **Phrasings from a finite set:** 108 impolite and 54 polite templates. The model learns these frames very well and has trouble with other formulations — the main reason for the gap between test and challenge.
- **Politeness metric:** a bag-of-words classifier trained on the same data; it recognises politeness markers, not genuine politeness, and saturates at ≈ 1.0 for every model.
- **Content metric:** word-based; it cannot see grammar errors or meaning reversals (hence the human evaluation).
- **Word-level vocabulary:** morphological variants are different tokens; tense and agreement changes are not learned.
- **One sentence (or two short ones), English only, at most 40 tokens.**
- The human evaluation was done by one annotator with AI assistance; it should be confirmed by more people.

9.4 Possible improvements

- More varied training data: paraphrase the rude openers and polite frames (more templates, or real sentences collected and checked by hand); add grammar-changing examples (*booked* → *book*).
- Subword tokenisation (BPE) so that *booked* and *book* share pieces.
- A Transformer encoder–decoder with the same copy mechanism, for comparison.
- A coverage penalty (See et al., 2017) against repetitions, and a better politeness classifier trained on independent data.
- A larger human evaluation with several annotators.

10. Conclusion

Main findings. A small LSTM encoder–decoder written from scratch learns to rewrite impolite sentences into polite ones. The attention mechanism and above all the copy mechanism are what make it keep the content: success on hand-written sentences rises from 11 % (no attention) to 22 % (attention) to 76 % (copy) by the automatic metric, 17 % → 53 % fully acceptable by human judgement. Making the model bigger, changing LSTM to GRU, or lowering teacher forcing did not help; making the data more diverse did.

Lessons learned. (1) Test on data that is *really* different from training: the in-domain test score (99.9 %) hid the real weakness. (2) An automatic metric must be checked against human judgement. (3) Hypotheses must be tested, and can fail: word dropout made the model worse, and we report it.

Future work. Collect real impolite messages (without insults) and their polite versions, switch to subword units, and compare with a Transformer.

Appendix — reproducing the numbers

`README.md` gives every command. Main files: `data/politeness_dataset.csv` (dataset), `checkpoints/C_copy_nodrop.pt` (final model), `reports/results.md` (all metrics), `reports/human_eval_*.csv` (human labels), `archive_v1/` (first dataset version).

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